

Task-Adaptive Semantic Communication for Multi-Task Inference Using Attention-Based Reasoning

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Abstract—The increasing reliance on autonomous and intelligent systems necessitates communication paradigms that move beyond data fidelity and explicitly support task-oriented reasoning under resource constraints. Semantic communication offers a promising alternative by prioritizing the transmission of meaning relevant to a downstream objective rather than raw sensory data. In this work, we present an end-to-end, goal-oriented semantic communication framework for multi-task traffic sign classification, where communication, reasoning, and inference are learned jointly. Using the German Traffic Sign Recognition Benchmark (GTSRB) dataset, a shared convolutional neural network (CNN) encoder maps input images into a 128-dimensional semantic representation that captures high-level attributes. A dynamic, task-adaptive attention-based reasoning module then identifies and selects a task-relevant subset of semantic features corresponding to different objectives, including shape, color, and speed-limit classification. The resulting compact semantic message is transmitted over a simulated additive white Gaussian noise (AWGN) channel and decoded directly for inference using lightweight task-specific multi-layer perceptron (MLP) decoders. The framework is trained in a multi-task learning setting and evaluated by varying the semantic message length from 128 down to 8 features and the signal-to-noise ratio (SNR) from -40 dB to 20 dB, thereby explicitly modeling inference under semantic sparsity and channel uncertainty. Experimental results demonstrate that the proposed system maintains strong task performance across all objectives even under severe compression and low-SNR conditions, indicating that task-adaptive semantic reasoning enables robust and efficient inference. These findings underscore the potential of learning-based semantic abstraction and reasoning mechanisms as key enablers of intelligent communication in next-generation autonomous networks.

Index Terms—Semantic communication, task-adaptive reasoning, semantic abstraction, multi-task learning, inference under uncertainty, intelligent systems, attention mechanisms.

I. INTRODUCTION

The rapid evolution of wireless communication technologies, together with the widespread deployment of autonomous and intelligent systems, has significantly increased the demand for communication frameworks that are both bandwidth-

efficient and resilient to channel impairments [1], [2]. Emerging applications such as autonomous driving, intelligent transportation systems, and cyber-physical systems increasingly rely on the timely exchange of high-dimensional perceptual data, including images, videos, and sensor measurements. In such scenarios, conventional communication systems, which are designed to ensure symbol-level fidelity, often incur excessive bandwidth consumption and exhibit limited robustness under adverse channel conditions [3].

Semantic communication has recently emerged as a promising paradigm that departs from traditional symbol-centric transmission by focusing on the reliable exchange of task-relevant meaning [4]–[6]. By aligning communication objectives with downstream inference or decision-making tasks, semantic communication enables goal-oriented and resource-efficient communication. This paradigm is particularly well suited for data-intensive modalities, where transmitting all available information is neither necessary nor optimal for achieving the desired task outcome [7].

Motivated by these advantages, recent studies have leveraged advances in deep learning to develop semantic communication frameworks for a wide range of data modalities, including text [8], speech [9], images [10], [11], and videos [12], [13]. In such learning-based semantic communication systems, neural networks are trained to extract and reconstruct semantic content, with latent feature embeddings serving as compact representations of meaning. However, these latent representations are inherently tied to specific network architectures and training objectives, leading to variations in structure, scale, and interpretability across different models. As a result, semantic information lacks a unified and transparent representation, which limits the generalization of semantic communication systems across heterogeneous tasks and application scenarios.

Despite its potential, several fundamental challenges continue to hinder the practical deployment of semantic communication systems. First, the absence of standardized and task-agnostic evaluation metrics complicates the comparison and benchmarking of existing approaches [14]. Second, many proposed systems remain tightly coupled to specific data modalities, such as text or images, which restricts their adaptability across diverse applications [15], [16]. Third, and more critically, most existing semantic communication frameworks are designed for single-task operation, offering limited support for scenarios in which multiple semantic objectives must be addressed simultaneously [17], [18]. In real-world applications such as traffic sign understanding, autonomous agents must concurrently reason about multiple attributes, including shape,

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color, and symbolic content, rendering single-task semantic communication strategies inefficient and inflexible.

To address issues related to interpretability and modality dependence, the Semantic Communication with Conceptual Spaces (SCCS) framework has been proposed as a principled approach for representing meaning in a geometric and task-independent manner [19]–[21]. By organizing semantic information into structured domains, SCCS facilitates interpretable abstraction and reasoning across tasks. However, prior SCCS-based implementations have largely relied on deterministic autoencoder architectures with manually tuned regularization terms, which limits scalability and robustness under uncertainty [19]–[21]. Recent advances suggest that probabilistic latent-variable models and distribution-aware learning objectives can significantly enhance semantic representation quality and generalization [22].

Beyond semantic representation, reasoning plays a central role in enabling intelligent semantic communication systems. Many existing learning-based approaches rely primarily on correlation-driven feature selection mechanisms, which implicitly learn statistical associations between input features and task objectives while overlooking deeper semantic dependencies [23]–[27]. Consequently, such systems may transmit redundant or weakly informative semantic features, resulting in suboptimal communication efficiency under bandwidth and noise constraints. Recent studies have emphasized that incorporating explicit reasoning mechanisms, supported by structured knowledge or learned semantic dependencies, can significantly improve task relevance, robustness, and interpretability in semantic communication systems [23]. At a broader level, emerging research on AI-native wireless networks further highlights the limitations of purely data-driven models and motivates the integration of reasoning- and causality-aware intelligence to enable adaptive, goal-oriented communication in dynamic environments [28]. Moreover, recent studies have shown that attention mechanisms can effectively prioritize task-relevant semantic information for transmission, thereby improving robustness and communication efficiency under resource constraints. In particular, attention-driven semantic communication schemes leverage importance estimation to selectively emphasize critical semantic components, enabling unequal protection and improved meaning preservation over noisy channels [29]. Attention-based designs have also been explored in AI-native wireless systems to guide semantic-aware transmission without requiring explicit task-specific supervision [30].

Motivated by these observations, this work focuses on learning task-adaptive semantic abstraction policies within a multi-task semantic communication framework, where attention plays a central role in semantic reasoning and feature selection. We adopt a multi-task learning (MTL) paradigm in which a shared convolutional neural network (CNN) encoder extracts a common semantic representation, and a dynamic attention-based reasoning module explicitly learns to assign importance weights to semantic features based on the task objective. By leveraging attention to selectively activate and transmit only task-relevant semantic components, the proposed framework enables fine-grained semantic prioritization and

efficient reuse of shared representations, while preserving task-specific discriminative power. In summary, this paper proposes a unified, goal-oriented semantic communication framework that integrates attention-driven semantic abstraction, shared representation learning, and multi-task inference within a single end-to-end architecture, enabling efficient, scalable, and interpretable multi-task inference under realistic communication constraints. The main contributions of this work are summarized as follows:

- We propose an end-to-end, goal-oriented semantic communication framework for multi-task inference, in which semantic abstraction, task-aware reasoning, and inference are jointly learned under communication constraints.
- We develop a task-adaptive attention-based reasoning mechanism that dynamically selects task-relevant semantic features from a shared latent representation, enabling efficient semantic abstraction without retraining for different tasks.
- We formulate and evaluate a multi-task semantic communication system that explicitly models inference under semantic sparsity and channel uncertainty by varying semantic message length and signal-to-noise ratio during training and evaluation.
- Through extensive experiments on the GTSRB dataset, we demonstrate that the proposed framework achieves robust multi-task performance under severe bandwidth reduction and low-SNR conditions, highlighting the effectiveness of task-adaptive semantic reasoning for intelligent communication systems.

The remainder of this paper is organized as follows. Section II introduces the proposed system model for multi-task semantic communication, including semantic encoding, task-adaptive reasoning, channel modeling, and task-specific decoding. Section III presents the end-to-end learning and inference framework, detailing the multi-task training objective, semantic abstraction strategy, and training under channel-induced uncertainty, along with the complete training and evaluation procedure. Section IV reports the simulation setup and experimental results, providing a comprehensive performance analysis under varying semantic message lengths and channel conditions. Finally, Section V concludes the paper and outlines directions for future research.

II. SYSTEM MODEL

A. System-Level Problem Formulation

We consider a goal-oriented semantic communication system designed to support multiple downstream tasks under communication constraints. Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote the input observation and let u_j denote the ground-truth label associated with task $j \in \{1, \dots, N_{\text{tasks}}\}$. The system aims to learn a task-adaptive semantic abstraction and inference strategy that minimizes task-specific prediction error under a communication budget constraint. Formally, the joint optimization problem can be expressed as

$$\min_{\theta_{\text{enc}}, \theta_{\text{attn}}, \{\theta_{\text{dec}, j}\}} \mathbb{E}_{(\mathbf{x}, u_j)} \left[\sum_{j=1}^{N_{\text{tasks}}} \ell_j(f_{\text{dec}, j}(\hat{\mathbf{z}}_{R, j}), u_j) \right], \quad (1)$$

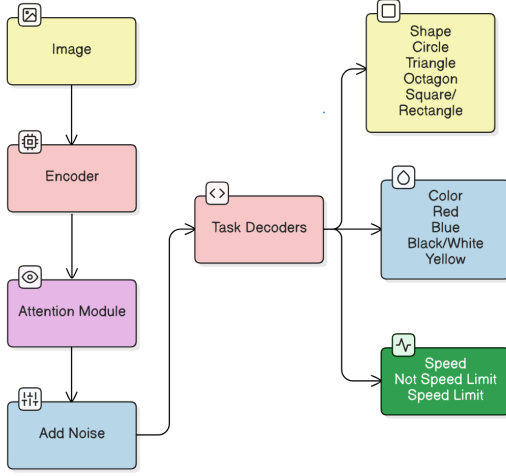


Fig. 1. System model of the proposed multi-task semantic communication framework. An input image is encoded into a shared semantic representation, which is processed by a task-adaptive attention module to select task-relevant semantic features. The selected semantic representation is transmitted over a noisy channel and decoded by task-specific decoders to infer shape, color, and speed-limit attributes.

subject to the semantic communication constraint

$$\|\mathbf{z}_{R,j}\|_0 \leq k, \quad (2)$$

where $\ell_j(\cdot)$ denotes the cross-entropy loss for task j , $\|\cdot\|_0$ represents the ℓ_0 sparsity measure, and k controls the semantic message length, i.e., the number of transmitted semantic dimensions. The system follows a transmitter–channel–receiver architecture, where semantic abstraction and task-aware reasoning are explicitly incorporated at the transmitter to enable efficient and intelligent communication. An overview of the system is shown in Fig. 1. At a high level, the transmitter extracts a shared semantic representation from the input data and subsequently applies a task-conditioned reasoning mechanism to identify and transmit only the most relevant semantic components. The resulting compact semantic representation is conveyed over a noisy channel to the receiver, which performs task-specific inference using lightweight decoders.

B. Input Representation and Semantic Encoding

We consider a goal-oriented semantic communication scenario in which high-dimensional perceptual data must be communicated efficiently to support multiple downstream inference tasks. Let the input to the system be an image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C denote the height, width, and number of channels of the image, respectively. Rather than transmitting the raw input $\mathbf{x}$, the system first maps the observation into a compact semantic representation that captures task-relevant information while suppressing redundant low-level details. This semantic abstraction is performed by a shared semantic encoder implemented as a convolutional neural network (CNN). The encoder is modeled as a parametric mapping

$$\mathbf{z} = f_{\text{enc}}(\mathbf{x}; \boldsymbol{\theta}_{\text{enc}}), \quad (3)$$

where $f_{\text{enc}}(\cdot)$ denotes the encoding function parameterized by $\boldsymbol{\theta}_{\text{enc}}$, and $\mathbf{z} \in \mathbb{R}^d$ is the resulting latent semantic representation. The dimension d controls the expressive capacity of the semantic space; throughout this work, we set $d = 128$.

The vector $\mathbf{z}$ is designed to act as a *shared semantic space* across all downstream tasks. Through multi-task training, the encoder learns to organize latent dimensions such that they jointly encode high-level attributes related to shape, color, and symbolic content. Importantly, no task-specific assumptions are imposed at this stage. Instead, the encoder is encouraged to learn a general-purpose semantic representation that can later be selectively exploited depending on the task objective.

From an information-processing perspective, the semantic encoder performs a dimensionality reduction that transforms the raw observation $\mathbf{x}$ into a structured representation $\mathbf{z}$ that is more amenable to reasoning and communication. This separation between semantic abstraction and task-specific decision-making is a key design principle of the proposed framework, as it allows the system to decouple *what is perceived* from *what is communicated* under task and bandwidth constraints. From an information-processing perspective, the encoder can be interpreted as a nonlinear mapping that compresses the high-dimensional observation $\mathbf{x}$ into a lower-dimensional semantic manifold $\mathbb{R}^d$. The learned latent vector $\mathbf{z}$ serves as a shared semantic space that preserves task-relevant information while suppressing redundant variability in the input domain.

C. Task-Adaptive Semantic Reasoning

While the shared semantic representation $\mathbf{z} \in \mathbb{R}^d$ captures a broad range of meaningful attributes present in the input observation, not all semantic components are equally relevant for every downstream task. In goal-oriented communication scenarios, transmitting the entire semantic vector is often unnecessary and inefficient, particularly under bandwidth and noise constraints. To address this challenge, we introduce a task-adaptive semantic reasoning module that selectively identifies and conveys only the semantic components that are most informative for a given task under a communication budget constraint.

Let $j \in \{1, \dots, N_{\text{tasks}}\}$ denote the index of the task to be performed at the receiver. Each task is associated with a distinct semantic objective and therefore requires a different subset of the latent representation. The role of the reasoning module is to construct a task-conditioned semantic projection of $\mathbf{z}$ that preserves task-relevant information while satisfying a sparsity constraint. Moreover, task awareness is introduced through a learnable task query vector $\mathbf{q}_j \in \mathbb{R}^d$, drawn from a task embedding matrix $\mathbf{T} \in \mathbb{R}^{N_{\text{tasks}} \times d}$. The task query encodes the semantic intent of task j and enables conditioning of the abstraction process on the task objective. Using an attention-based interaction between the task query and the semantic representation, the reasoning module computes a relevance score for each latent dimension:

$$s_j = g_{\text{attn}}(\mathbf{q}_j, \mathbf{z}; \boldsymbol{\theta}_{\text{attn}}), \quad (4)$$

where $s_j \in [0, 1]^d$ denotes the vector of semantic importance scores and $\boldsymbol{\theta}_{\text{attn}}$ represents the parameters of the attention

mechanism. Each element $s_{j,i}$ quantifies the estimated contribution of the i -th latent dimension to task j . To enforce the communication constraint, we define a Top- k projection operator $\mathcal{T}_k(\cdot)$ that retains the k largest elements of its argument and sets all remaining components to zero. The task-specific binary mask is therefore expressed compactly as

$$\mathbf{m}_j = \mathcal{T}_k(\mathbf{s}_j), \quad (5)$$

where $\mathbf{m}_j \in \{0, 1\}^d$ satisfies $\|\mathbf{m}_j\|_0 = k$. The task-adaptive semantic message is then obtained via a diagonal selection operator:

$$\mathbf{z}_{R,j} = \text{diag}(\mathbf{m}_j)\mathbf{z}, \quad (6)$$

where $\mathbf{z}_{R,j} \in \mathbb{R}^d$ denotes the sparsified semantic representation prepared for transmission and $\|\mathbf{z}_{R,j}\|_0 \leq k$. To quantify communication efficiency, we define the normalized semantic rate

$$R_s = \frac{k}{d}, \quad (7)$$

where d is the latent dimension and k is the number of transmitted semantic features. The semantic rate $R_s \in (0, 1]$ measures the fraction of semantic dimensions conveyed to the receiver and explicitly captures the trade-off between communication cost and representational richness. Moreover, from an information-theoretic perspective, the reasoning module can be interpreted as learning a task-conditioned projection operator that seeks to maximize the mutual information between the transmitted representation and the task label under a sparsity constraint:

$$\max_{\mathbf{m}_j} I(u_j; \mathbf{z}_{R,j}) \quad \text{subject to} \quad \|\mathbf{z}_{R,j}\|_0 \leq k. \quad (8)$$

Thus, the semantic abstraction process aims to preserve task-relevant information while respecting the communication budget. Overall, the reasoning module implements a learned semantic abstraction policy

$$\Pi_j : \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad \Pi_j(\mathbf{z}) = \mathbf{z}_{R,j},$$

which maps the shared semantic representation to a compact task-specific message. Unlike static compression schemes, this projection adapts dynamically to both the input content and the task objective, enabling efficient multi-task semantic communication under identical bandwidth constraints.

D. Communication Channel Model

Once the task-adaptive semantic representation $\mathbf{z}_{R,j}$ is obtained, it is transmitted over a noisy communication channel from the transmitter to the receiver. In the context of semantic communication, the channel introduces uncertainty that perturbs the conveyed semantic evidence and directly impacts the reliability of downstream task inference. A probabilistic characterization of this uncertainty is therefore essential for analyzing robustness and performance trade-offs. In this work, the communication medium is modeled as an additive white Gaussian noise (AWGN) channel. The received semantic representation is given by

$$\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w}, \quad (9)$$

where $\hat{\mathbf{z}}_{R,j} \in \mathbb{R}^d$ denotes the received semantic vector and $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$ represents additive Gaussian noise with zero

mean and covariance matrix $\sigma_n^2 \mathbf{I}$. Here, $\mathbf{I}$ denotes the $d \times d$ identity matrix and σ_n^2 is the noise variance per semantic dimension. Let

$$P_s = \mathbb{E}[\|\mathbf{z}_{R,j}\|^2] \quad (10)$$

denote the average transmit power of the semantic message, where the expectation is taken over the data distribution. For a given linear signal-to-noise ratio SNR, the noise variance is defined as

$$\sigma_n^2 = \frac{P_s}{\text{SNR}}. \quad (11)$$

Due to the sparsification induced by the reasoning module, $\mathbf{z}_{R,j}$ contains only k non-zero semantic components. Consequently, the effective signal energy is concentrated on task-relevant dimensions, and the channel noise perturbs only those transmitted components. This establishes an intrinsic interaction between semantic compression (controlled by k) and channel uncertainty (controlled by SNR). From a probabilistic inference perspective, the receiver observes a noisy semantic vector $\hat{\mathbf{z}}_{R,j}$ and performs task inference under uncertainty. The optimal decision rule for task j can be expressed as maximum a posteriori (MAP) estimation:

$$\hat{u}_j = \arg \max_{c \in \{1, \dots, C_j\}} P(u_j = c \mid \hat{\mathbf{z}}_{R,j}), \quad (12)$$

where $\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w}$. Thus, the task-specific decoder implicitly approximates Bayesian inference over noisy semantic evidence. To characterize performance under joint compression and channel effects, we define the task-specific semantic risk as

$$\mathcal{R}_j(k, \text{SNR}) = \mathbb{E}[\ell_j(f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j}), u_j)], \quad (13)$$

which explicitly depends on both the semantic message length k and the channel quality. This formulation reveals the fundamental trade-off underlying semantic communication: reducing k improves communication efficiency but may increase inference risk, particularly under low SNR conditions. Accordingly, the received semantic representation $\hat{\mathbf{z}}_{R,j}$ can be interpreted as a noisy and partially observed projection of the underlying task-relevant semantics. The decoder must therefore learn robust decision boundaries that compensate for both sparsity-induced information loss and channel-induced perturbations, enabling reliable goal-oriented inference without reconstructing the original input signal.

E. Task-Specific Semantic Decoding

At the receiver, the objective is to infer the desired task outcome from the received semantic representation $\hat{\mathbf{z}}_{R,j}$, which is both sparsified and corrupted by channel noise. Unlike conventional communication systems that aim to reconstruct the transmitted signal, the proposed framework directly performs goal-oriented inference from noisy semantic evidence. This shift from signal recovery to decision-centric inference is a defining characteristic of semantic communication. For each task $j \in \{1, \dots, N_{\text{tasks}}\}$, a dedicated semantic decoder maps the received semantic vector $\hat{\mathbf{z}}_{R,j} \in \mathbb{R}^d$ to a task-specific decision space. The decoder is implemented as a parametric function

$$f_{\text{dec},j} : \mathbb{R}^d \rightarrow \mathbb{R}^{C_j},$$

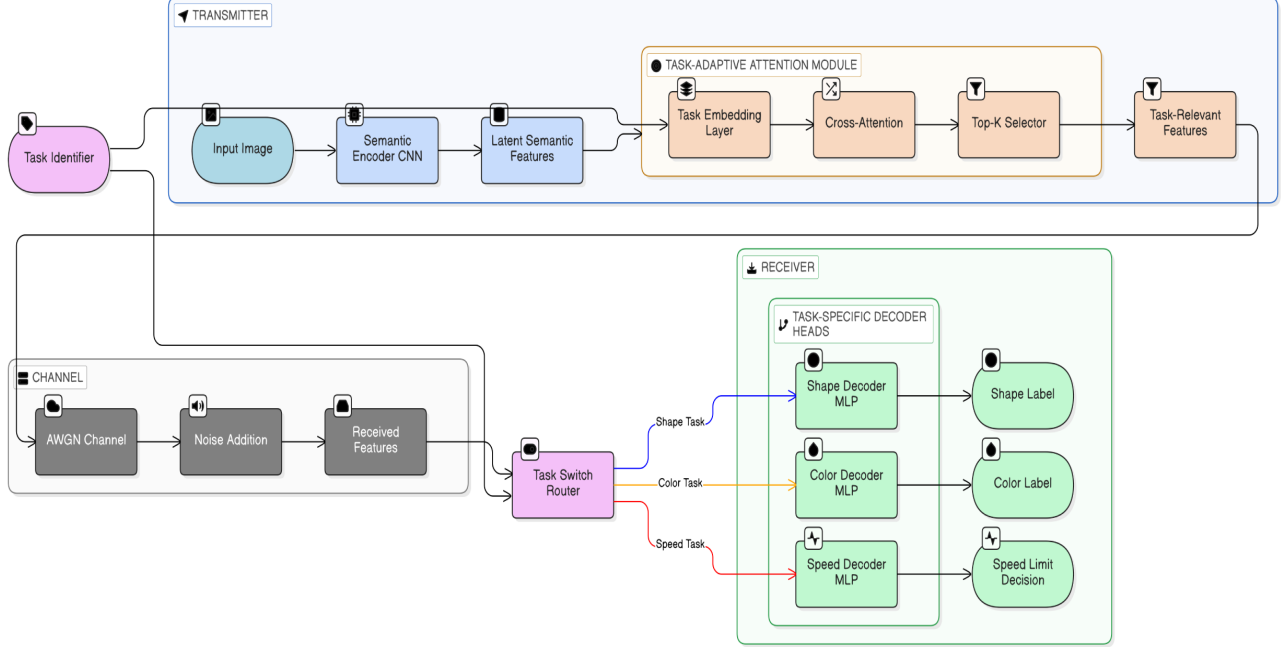


Fig. 2. End-to-end architecture of the proposed task-adaptive semantic communication framework. The transmitter maps the input image to a latent semantic representation and applies a task-conditioned projection with Top-K selection to generate a compact semantic message, which is transmitted over an AWGN channel. The receiver performs probabilistic task inference directly from the received semantic representation. All components are jointly optimized under communication constraints.

with parameters $\theta_{\text{dec},j}$. The decoding operation is expressed as

$$\mathbf{y}_j = f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j}; \theta_{\text{dec},j}), \quad (14)$$

where $\mathbf{y}_j \in \mathbb{R}^{C_j}$ denotes the vector of output logits and C_j is the number of classes associated with task j . To interpret the logits probabilistically, a softmax mapping defines the posterior distribution over task labels:

$$P(u_j = c \mid \hat{\mathbf{z}}_{R,j}) = \frac{\exp([\mathbf{y}_j]_c)}{\sum_{c'=1}^{C_j} \exp([\mathbf{y}_j]_{c'})}. \quad (15)$$

The final task estimate is obtained via maximum a posteriori (MAP) decision:

$$\hat{u}_j = \arg \max_{c \in \{1, \dots, C_j\}} P(u_j = c \mid \hat{\mathbf{z}}_{R,j}). \quad (16)$$

Under this formulation, the decoder implements an approximation of Bayesian inference over noisy semantic observations, where uncertainty arises from both channel perturbations and sparsity-induced information loss. The task-specific decoders are trained jointly with the semantic encoder and the reasoning module in an end-to-end manner. During training, the decoders are exposed to semantic representations characterized by varying sparsity levels k and channel conditions (SNR), enabling them to learn robust decision boundaries under partial observability. In conjunction with the semantic risk defined earlier,

$$\mathcal{R}_j(k, \text{SNR}) = \mathbb{E} [\ell_j(f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j}), u_j)], \quad (17)$$

the decoder serves as the final decision operator that minimizes task-specific inference error given a compressed and noisy semantic message. From a structural perspective, the complete

receiver operation can be interpreted as the composition

$$\hat{u}_j = f_{\text{dec},j}(\Pi_j(\mathbf{z}) + \mathbf{w}),$$

where $\Pi_j(\mathbf{z}) = \mathbf{z}_{R,j}$ denotes the task-conditioned semantic projection learned at the transmitter. This decomposition clarifies that the overall system learns a family of task-adaptive projections followed by probabilistic decision mappings, thereby separating semantic abstraction from inference. This modular decoding strategy allows the receiver to support multiple tasks using lightweight models while relying on a shared semantic abstraction and reasoning process at the transmitter. By decoupling semantic representation learning, task-adaptive projection, and probabilistic inference, the proposed system achieves flexibility, robustness, and scalability in multi-task semantic communication scenarios.

F. System Architecture Overview

Fig. 2 illustrates the complete end-to-end architecture of the proposed task-adaptive semantic communication framework. The system follows a transmitter–channel–receiver paradigm that explicitly decomposes the overall mapping

$$\mathbf{x} \longrightarrow \hat{u}_j$$

into semantic encoding, task-conditioned projection, noisy transmission, and probabilistic inference. At the transmitter, a task identifier specifying the semantic objective j is provided. The input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ is first mapped by the shared encoder

$$f_{\text{enc}} : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^d$$

to produce the latent semantic representation $\mathbf{z} \in \mathbb{R}^d$.

The reasoning module then implements the learned task-conditioned projection

$$\Pi_j(\mathbf{z}) = \mathbf{z}_{R,j},$$

where $\Pi_j : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is realized through task embedding, cross-attention, and Top- k projection. The resulting semantic message $\mathbf{z}_{R,j}$ satisfies the sparsity constraint $\|\mathbf{z}_{R,j}\|_0 \leq k$, corresponding to a semantic rate $R_s = k/d$. The compact semantic message is transmitted over the AWGN channel:

$$\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I}).$$

At the receiver, a task switch routes the received semantic vector to the appropriate task-specific decoder

$$f_{\text{dec},j} : \mathbb{R}^d \rightarrow \mathbb{R}^{C_j},$$

which computes logits and performs MAP decision:

$$\hat{u}_j = \arg \max_c P(u_j = c \mid \hat{\mathbf{z}}_{R,j}).$$

Unlike conventional communication systems that reconstruct $\mathbf{x}$, the proposed framework directly estimates the task output $\hat{u}_j$ from compressed and noisy semantic evidence. The architecture therefore implements the composite mapping

$$\hat{u}_j = f_{\text{dec},j}(\Pi_j(f_{\text{enc}}(\mathbf{x})) + \mathbf{w}),$$

which clarifies the separation between semantic abstraction, communication, and decision-making. All components depicted in Fig. 2 are trained jointly to minimize the semantic risk

$$\mathcal{R}_j(k, \text{SNR}) = \mathbb{E} [\ell_j(f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j}), u_j)],$$

thereby enabling coordinated learning of representation, task-adaptive projection, and inference under communication constraints. The modular structure allows extension to additional tasks, alternative projection mechanisms, or more realistic channel models while preserving the core objective of maximizing task-relevant information transfer under limited semantic rate.

Unified Optimization Formulation

Theorem 1 (Stochastic Constrained Semantic Risk Minimization). Consider a multi-task semantic communication system with N_{tasks} downstream tasks. Let the semantic encoder $f_{\text{enc}} : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^d$ produce a shared latent representation $\mathbf{z} = f_{\text{enc}}(\mathbf{x}; \boldsymbol{\theta}_{\text{enc}})$. For each task j , let the task-adaptive projection operator $\Pi_j(\cdot; \boldsymbol{\theta}_{\text{attn}})$ generate a sparse semantic message

$$\mathbf{z}_{R,j} = \Pi_j(\mathbf{z}), \quad \|\mathbf{z}_{R,j}\|_0 \leq k,$$

which is transmitted over an AWGN channel,

$$\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I}).$$

Let $f_{\text{dec},j}(\cdot; \boldsymbol{\theta}_{\text{dec},j})$ denote the task-specific decoder that produces the posterior $P(u_j \mid \hat{\mathbf{z}}_{R,j})$ and let $\ell_j(\cdot)$ denote the cross-entropy loss for task j . Then the end-to-end learning problem is equivalent to solving the following stochastic constrained optimization:

$$\min_{\boldsymbol{\theta}_{\text{enc}}, \boldsymbol{\theta}_{\text{attn}}, \{\boldsymbol{\theta}_{\text{dec},j}\}} \sum_{j=1}^{N_{\text{tasks}}} \mathbb{E}_{k, \text{SNR}} [\mathcal{R}_j(k, \text{SNR})]$$

subject to the semantic rate constraint

$$\mathbb{E}[\|\Pi_j(\mathbf{z})\|_0] \leq k, \quad R_s = \frac{k}{d},$$

where the task-specific semantic risk is defined as

$$\mathcal{R}_j(k, \text{SNR}) = \mathbb{E}_{(\mathbf{x}, u_j)} [\ell_j(f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j}), u_j)].$$

Under this formulation, the learned projection operator Π_j implicitly seeks to maximize task-relevant mutual information $I(u_j; \mathbf{z}_{R,j})$ while respecting the semantic rate constraint. ■

Remark 1 (Semantic Rate–Risk Trade-off). The optimization in Theorem 1 reveals an intrinsic trade-off between semantic communication efficiency and inference reliability. Let $R_s = k/d$ denote the normalized semantic rate and $\mathcal{R}_j(k, \text{SNR})$ denote the task-specific semantic risk. Then, for fixed channel conditions,

$$\frac{\partial \mathcal{R}_j}{\partial k} \leq 0,$$

indicating that increasing the semantic message length generally reduces task inference risk by preserving more task-relevant information. Similarly, for fixed semantic rate,

$$\frac{\partial \mathcal{R}_j}{\partial \text{SNR}} \leq 0,$$

implying improved robustness as channel quality increases.

However, increasing k also increases communication cost, as $R_s = k/d$. Consequently, semantic communication design reduces to balancing semantic rate and semantic risk, analogous to the rate–distortion trade-off in classical information theory. In contrast to distortion minimization, however, the objective here is goal-oriented risk minimization, where only task-relevant information must be preserved under bandwidth and channel constraints.

III. END-TO-END LEARNING AND INFERENCE FRAMEWORK

The proposed framework is trained and evaluated through a unified learning process that jointly optimizes semantic representation learning, task-adaptive reasoning, and task-specific inference. Unlike conventional communication systems that separate feature extraction, compression, and decoding, the proposed approach learns these components end-to-end with the explicit objective of enabling reliable multi-task inference from compact semantic representations transmitted over a noisy channel. The working process is designed to reflect the fundamental principles of semantic communication: abstraction of meaning, task-awareness, and robustness to uncertainty. In particular, the training procedure encourages the system to learn which semantic components are causally and functionally relevant for each task, and how to utilize these components effectively under varying communication constraints.

A. End-to-End Multi-Task Learning Objective

Let $j \in \{1, \dots, N_{\text{tasks}}\}$ denote the index of the downstream tasks supported by the system. For a given input image $\mathbf{x}$, the encoder produces a shared semantic representation

$$\mathbf{z} = f_{\text{enc}}(\mathbf{x}; \boldsymbol{\theta}_{\text{enc}}) \in \mathbb{R}^d,$$

which is transformed by the task-conditioned projection operator

$$\mathbf{z}_{R,j} = \Pi_j(\mathbf{z}; \boldsymbol{\theta}_{\text{attn}})$$

into a sparse semantic message satisfying $\|\mathbf{z}_{R,j}\|_0 \leq k$. After transmission over the AWGN channel,

$$\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w},$$

the task-specific decoder produces a probabilistic prediction

$$P(u_j = c \mid \hat{\mathbf{z}}_{R,j}) = \text{Softmax}(f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j}; \boldsymbol{\theta}_{\text{dec},j})).$$

The training objective is formulated as minimization of the expected semantic risk across all tasks:

$$\mathcal{L}_{\text{total}} = \sum_{j=1}^{N_{\text{tasks}}} \mathbb{E}_{(\mathbf{x}, u_j)} [\ell_j(f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j}), u_j)], \quad (18)$$

where $\ell_j(\cdot)$ denotes the categorical cross-entropy loss for task j . In empirical form, over a dataset of size N , the objective becomes

$$\mathcal{L}_{\text{total}} = \sum_{j=1}^{N_{\text{tasks}}} \frac{1}{N} \sum_{n=1}^N \mathcal{L}_{\text{CE}}(u_j^{(n)}, \hat{u}_j^{(n)}). \quad (19)$$

End-to-end optimization jointly updates the encoder parameters $\boldsymbol{\theta}_{\text{enc}}$, the projection parameters $\boldsymbol{\theta}_{\text{attn}}$, and the task-specific decoder parameters $\boldsymbol{\theta}_{\text{dec},j}$ according to

$$\min_{\boldsymbol{\theta}_{\text{enc}}, \boldsymbol{\theta}_{\text{attn}}, \{\boldsymbol{\theta}_{\text{dec},j}\}} \mathcal{L}_{\text{total}}, \quad \text{subject to } \|\mathbf{z}_{R,j}\|_0 \leq k.$$

This formulation makes explicit that the learning process simultaneously structures the shared semantic space $\mathbf{z}$, learns task-conditioned projection operators Π_j that preserve task-relevant information under a semantic rate constraint $R_s = k/d$, and trains the decoders to approximate Bayesian inference under sparsity and channel uncertainty. Task specialization therefore emerges implicitly through joint optimization of abstraction and inference, rather than through task-specific supervision at the encoder level.

B. Learning Task-Adaptive Semantic Abstraction

A central component of the end-to-end learning process is the optimization of the task-conditioned projection operators Π_j , which map the shared semantic representation $\mathbf{z}$ to sparse task-specific messages $\mathbf{z}_{R,j}$. During training, the full semantic vector $\mathbf{z} \in \mathbb{R}^d$ is not transmitted. Instead, for each task j , the reasoning module produces a sparsified representation

$$\mathbf{z}_{R,j} = \Pi_j(\mathbf{z}), \quad \|\mathbf{z}_{R,j}\|_0 \leq k,$$

where k denotes the semantic message length and determines the semantic rate $R_s = k/d$. To promote robustness across communication regimes, the semantic message length k is not fixed during training. Instead, k is sampled from a predefined set of admissible sparsity levels, exposing the system to varying semantic rates. Similarly, the channel signal-to-noise ratio is varied during training, yielding noisy observations

$$\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I}),$$

where σ_n^2 is determined by the selected signal-to-noise ratio.

Under this stochastic training regime, the expected semantic risk becomes

$$\mathbb{E}_{k, \text{SNR}} [\mathcal{R}_j(k, \text{SNR})],$$

Algorithm 1 End-to-End Training and Evaluation of the Task-Adaptive Semantic Communication Framework

Require: Training dataset $\{(\mathbf{x}^{(n)}, \{u_j^{(n)}\}_{j=1}^{N_{\text{tasks}}})\}_{n=1}^N$

Require: Admissible semantic message lengths $\mathcal{K}$, SNR set $\mathcal{S}$

```

1: Initialize parameters  $\boldsymbol{\theta}_{\text{enc}}, \boldsymbol{\theta}_{\text{attn}}, \{\boldsymbol{\theta}_{\text{dec},j}\}$ 
2: for each training epoch do
3:   for each mini-batch do
4:     Sample  $k \sim \mathcal{K}$  and  $\text{SNR} \sim \mathcal{S}$ 
5:     Compute  $\mathbf{z} = f_{\text{enc}}(\mathbf{x})$ 
6:     for each task  $j$  do
7:        $\mathbf{z}_{R,j} = \Pi_j(\mathbf{z})$  subject to  $\|\mathbf{z}_{R,j}\|_0 \leq k$ 
8:        $\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w}$ 
9:       Compute task loss  $\ell_j$ 
10:    end for
11:    Update parameters by minimizing  $\sum_j \ell_j$ 
12:  end for
13: end for
14: Evaluation:
15: for each task  $j$  do
16:   for each  $k \in \mathcal{K}$  and  $\text{SNR} \in \mathcal{S}$  do
17:     Evaluate accuracy and semantic risk  $\mathcal{R}_j(k, \text{SNR})$ 
18:   end for
19: end for
```

which encourages the learned projection operator Π_j to preserve semantic dimensions that consistently contribute to task performance across different sparsity levels and channel conditions. The complete stochastic optimization procedure implementing this semantic risk minimization is summarized in Algorithm 1. The algorithm formalizes the joint learning of the encoder parameters $\boldsymbol{\theta}_{\text{enc}}$, the task-adaptive reasoning parameters embedded in Π_j , and the task-specific decoder parameters $\boldsymbol{\theta}_{\text{dec},j}$ under randomly sampled semantic rates and channel realizations. By exposing the system to varying k and SNR values during training, the projection operators are optimized to remain stable across multiple communication regimes.

From an optimization perspective, this procedure prevents the reasoning module from overfitting to a fixed subset of latent dimensions. Instead, it promotes the emergence of a structured semantic space in which task-relevant information is distributed across dimensions in a manner that remains robust under varying compression levels and noise intensities. Consequently, different tasks learn to attend to distinct regions of the shared semantic manifold while maintaining resilience to channel-induced perturbations. Empirically, this behavior manifests in distinct attention patterns across tasks, even though all tasks share a common encoder. The resulting semantic abstraction strategy therefore generalizes across operating conditions without requiring retraining or manual tuning of communication parameters.

C. Training Under Channel-Induced Uncertainty

In addition to semantic sparsification, the proposed framework is explicitly trained under stochastic channel uncertainty

to promote robustness of semantic inference. After task-adaptive abstraction, the semantic message $\mathbf{z}_{R,j} = \Pi_j(\mathbf{z})$ is transmitted through an additive noise channel, yielding the received representation

$$\hat{\mathbf{z}}_{R,j} = \mathbf{z}_{R,j} + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I}),$$

where $\hat{\mathbf{z}}_{R,j} \in \mathbb{R}^d$ and σ_n^2 is determined by the operating signal-to-noise ratio (SNR). This corruption process models the uncertainty introduced by the physical communication channel and directly perturbs the task-relevant semantic evidence. During training, the SNR is randomly sampled from a pre-defined set of admissible channel conditions, as summarized in Algorithm 1. Consequently, the decoders receive semantic representations drawn from a family of noisy distributions

$$\hat{\mathbf{z}}_{R,j} \sim \mathcal{N}(\mathbf{z}_{R,j}, \sigma_n^2 \mathbf{I}),$$

with varying noise intensities. This stochastic exposure encourages the decoders to learn decision functions that are stable with respect to perturbations in the semantic space, rather than relying on precise latent values. Formally, for task j , the training objective under channel uncertainty can be interpreted as minimizing the expected task loss over the joint distribution of semantic sparsity and channel noise,

$$\mathbb{E}_{k, \text{SNR}} [\mathcal{L}_{\text{CE}}(u_j, \hat{u}_j)],$$

where $\hat{u}_j = f_{\text{dec},j}(\hat{\mathbf{z}}_{R,j})$ and the expectation is taken with respect to the stochastic sampling of k and SNR during training. By jointly varying semantic sparsity and channel conditions, the learning process approximates semantic risk minimization under partial observability. The decoders are therefore optimized to infer task outcomes from incomplete and noisy semantic evidence. This structured exposure to uncertainty explains the strong empirical performance observed at low SNRs and short message lengths, as the model is explicitly conditioned to operate reliably under constrained communication regimes.

D. End-to-End Training and Evaluation Procedure

Building on the learning principles described above, we now summarize the complete end-to-end training and evaluation procedure used to optimize the proposed semantic communication framework. The objective of this procedure is to jointly learn (i) a shared semantic representation, (ii) a task-adaptive semantic reasoning mechanism, and (iii) task-specific inference models, while explicitly accounting for semantic sparsification and channel-induced uncertainty.

During training, the system processes each input sample through the semantic encoder to obtain a shared latent representation. For each task, a task-adaptive semantic message is generated using the reasoning module, corrupted by channel noise, and decoded to produce a task prediction. The losses from all tasks are aggregated and used to update the parameters of all modules simultaneously. By randomizing both the semantic message length and the channel signal-to-noise ratio during training, the procedure ensures that the learned models remain robust across a wide range of communication conditions. The complete training and evaluation workflow is summarized in Algorithm 1. The algorithm highlights the

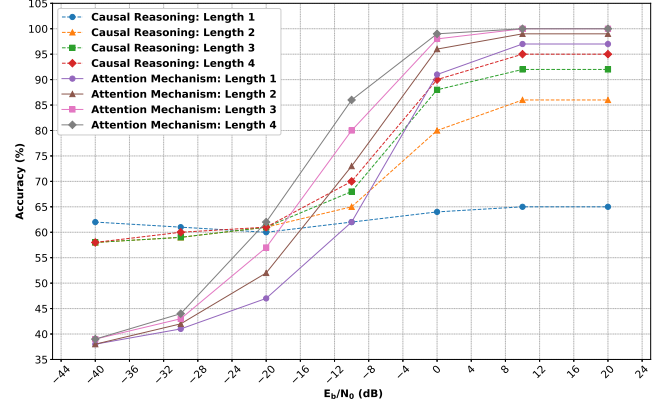


Fig. 3. Shape classification accuracy vs. SNR for different message lengths.

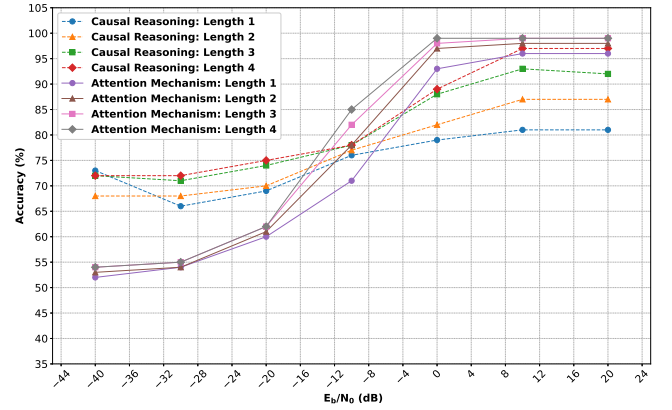


Fig. 4. Color classification accuracy vs. SNR for different message lengths.

interaction between semantic abstraction, task-aware reasoning, and inference under uncertainty, and directly reflects the working process underlying the experimental results presented in this paper.

E. Inference and Evaluation Logic

At inference time, the system operates without retraining. For a given task j , the reasoning module generates a task-adaptive semantic message of fixed length k , which is transmitted over a channel operating at a specified SNR. The receiver then performs task inference using the corresponding decoder.

Performance is evaluated by measuring task accuracy across different message lengths and SNR levels. This evaluation protocol reveals the trade-off between communication efficiency and inference performance and directly reflects the effectiveness of the learned semantic abstraction and reasoning mechanisms. In particular, the ability of the system to maintain high accuracy with very short semantic messages demonstrates that the reasoning module successfully identifies and preserves the most informative semantic components for each task.

IV. SIMULATION AND RESULTS

This section evaluates the proposed multi-task semantic communication framework and examines how task-adaptive

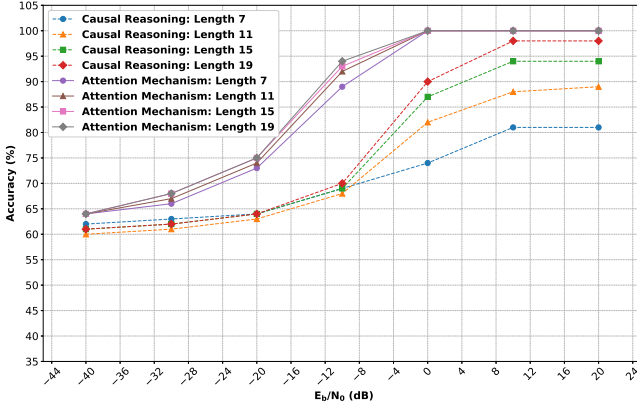


Fig. 5. Speed limit classification accuracy vs. SNR for different message lengths.

semantic reasoning influences inference performance under communication constraints. The system is trained for 15 epochs using the procedure described in Section IV and evaluated on a held-out test set. Performance is assessed across multiple dimensions, including task accuracy under varying channel conditions, robustness to semantic sparsification, interpretability of the learned reasoning mechanism, and discriminative ability under severe noise. Unless otherwise stated, results are reported as functions of the channel signal-to-noise ratio (SNR) and the semantic message length k , which denotes the number of retained semantic dimensions in the task-adaptive representation $\mathbf{z}_{R,j} \in \mathbb{R}^d$.

A. Task Accuracy Under Semantic and Channel Constraints

We first examine the impact of semantic abstraction and channel noise on task inference performance. Figs. 3–5 illustrate the classification accuracy as a function of SNR for different semantic message lengths across the three considered tasks. These results directly reflect the ability of the reasoning module to preserve task-relevant semantic information under bandwidth limitations. For the shape classification task, shown in Fig. 3, accuracy improves monotonically with increasing SNR and semantic message length. When the message length is small (e.g., $k = 8$), performance degrades more rapidly at low SNRs, indicating increased sensitivity to noise due to aggressive semantic sparsification. Nevertheless, the system maintains non-trivial accuracy even in this regime, demonstrating that the reasoning module consistently prioritizes semantic dimensions that capture the geometric structure of traffic signs.

As the message length increases, the performance gap between low and high SNR regimes narrows, suggesting that additional semantic dimensions provide redundancy that improves robustness to channel perturbations. This behavior is consistent with the training strategy described in Section IV, where the model is explicitly optimized to perform inference from incomplete and noisy semantic representations.

Fig. 4 presents the accuracy trends for the color classification task. Compared to shape classification, this task exhibits greater sensitivity to channel noise, particularly at shorter

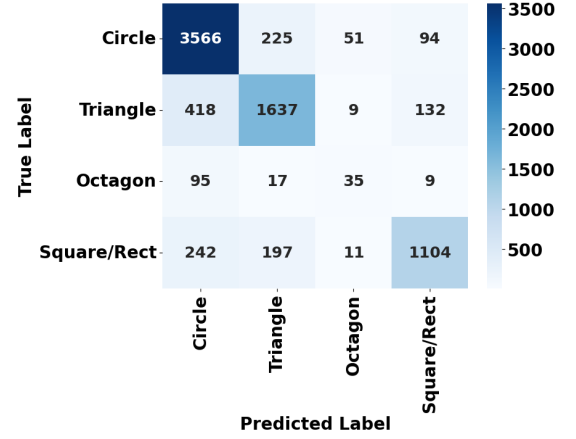


Fig. 6. Shape confusion matrix at SNR=10 dB, $L = 4$.

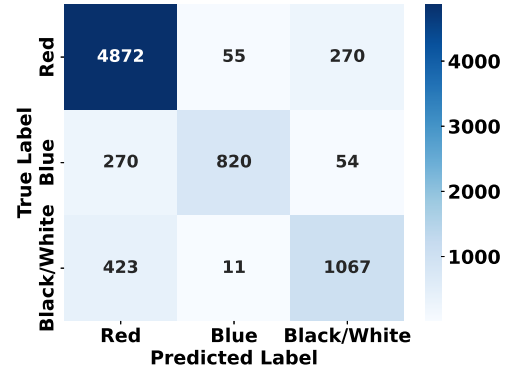


Fig. 7. Color confusion matrix at SNR=10 dB, $L = 4$.

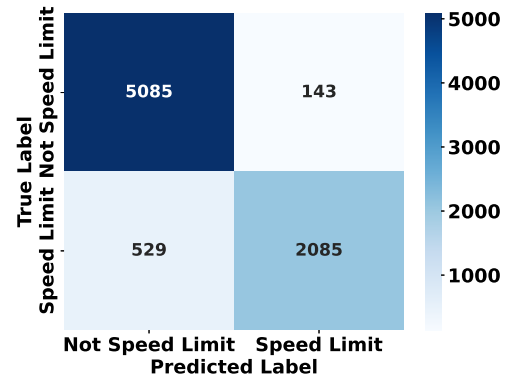


Fig. 8. Speed-limit confusion matrix at SNR=10 dB, $L = 4$.

message lengths. This observation reflects the fact that color-related semantics are often encoded in fewer latent dimensions and are therefore more vulnerable to corruption when aggressive sparsification is applied. Despite this sensitivity, the system achieves robust performance at moderate message lengths, indicating that the reasoning module successfully identifies and preserves the most discriminative color-related semantic features. These results highlight the inherent trade-off between communication efficiency and inference reliability

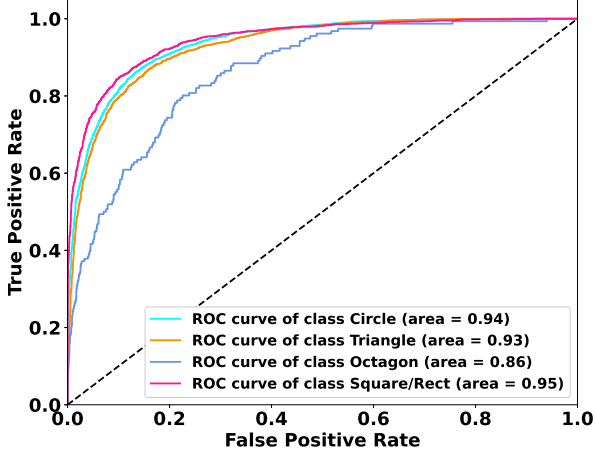


Fig. 9. ROC curve for Shape classification at SNR=-10 dB, $L = 4$.

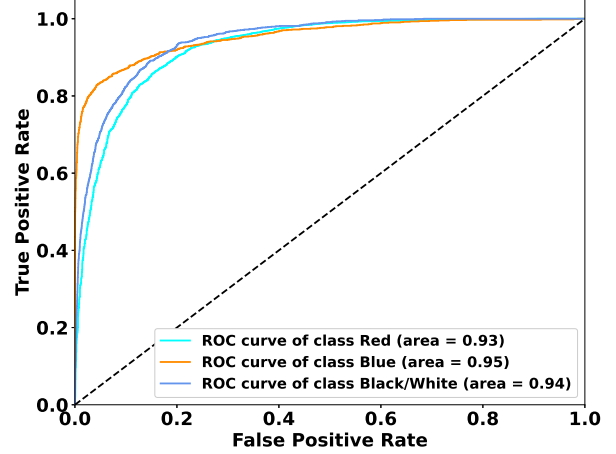


Fig. 10. ROC curve for Color classification at SNR=-10 dB, $L = 4$.

in semantic communication systems.

The binary speed-limit classification task, shown in Fig. 5, achieves high accuracy even for small message lengths and low SNR values. This behavior can be attributed to the relatively low semantic complexity of the task, which requires fewer semantic dimensions to support reliable inference. The strong performance in this regime demonstrates the flexibility of the proposed framework in adapting its abstraction strategy to tasks of varying complexity.

B. Classification Behavior Under Heavy Semantic Compression

To further analyze inference behavior under extreme communication constraints, confusion matrices are evaluated at an SNR of 10 dB using a semantic message length of $k = 4$. The corresponding confusion matrices for shape, color, and speed-limit classification are shown in Figs. 6–8.

Across all tasks, the confusion matrices exhibit strong diagonal dominance, indicating that the system retains high classification accuracy even when only a small number of semantic dimensions are transmitted. Misclassifications primarily occur between semantically similar classes, such as geometrically related shapes or visually similar colors, suggesting that errors arise from intrinsic semantic ambiguity rather than random noise.

C. Discriminative Ability Under Severe Noise

Finally, the discriminative capability of the proposed framework is evaluated using receiver operating characteristic (ROC) curves at an SNR of -10 dB with a semantic message length of $k = 4$. Figs. 9–11 show the ROC curves for the three tasks.

All tasks achieve strong discriminative performance, with area-under-the-curve (AUC) values ranging from 0.92 to 0.97. Notably, the speed-limit classification task attains an AUC of 0.97, while shape and color classification maintain robust

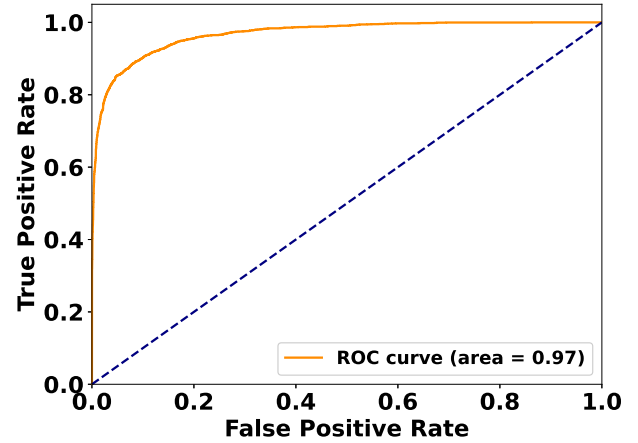


Fig. 11. ROC curve for Speed-limit classification at SNR=-10 dB, $L = 4$.

separation despite severe noise and aggressive semantic compression. These results confirm that the learned semantic representations and reasoning mechanisms enable reliable inference even under highly adverse communication conditions.

D. Interpretability of Task-Adaptive Semantic Reasoning

To gain insight into the learned reasoning behavior, Fig. 12 visualizes the semantic importance scores produced by the reasoning module for different input samples and tasks. The distributions exhibit clear task-dependent and input-dependent variations, indicating that the model does not rely on a fixed subset of semantic dimensions.

Instead, the reasoning module dynamically adapts its abstraction strategy based on both the semantic content of the input and the task objective. This behavior aligns with the conceptual design of the framework, where attention-based reasoning is employed to identify task-relevant semantic evidence rather than performing static feature selection.

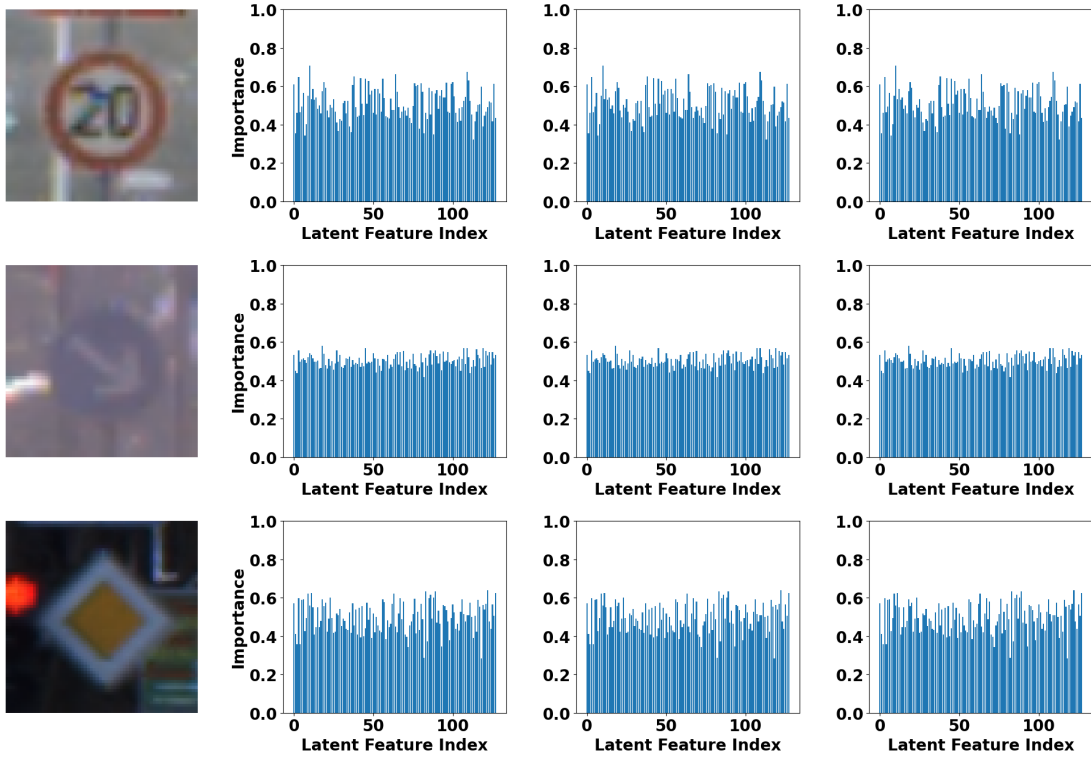


Fig. 12. Learned attention-based feature importance across tasks.

V. CONCLUSION AND FUTURE WORK

This work investigated a goal-oriented semantic communication framework for multi-task traffic sign classification, in which semantic abstraction, task-adaptive reasoning, and inference are learned jointly in an end-to-end manner. By integrating a shared CNN encoder with a dynamic attention-based reasoning module and lightweight task-specific decoders within a multi-task learning paradigm, the proposed system effectively decouples semantic understanding from raw data transmission. Experimental results demonstrate that selectively transmitting task-relevant semantic features enables reliable multi-task inference across a wide range of channel conditions while substantially reducing communication overhead. The observed robustness under severe semantic compression and low SNR conditions confirms that task-adaptive semantic reasoning provides an effective mechanism for balancing communication efficiency and inference performance in intelligent systems.

Future work will focus on enhancing the reasoning module using more expressive architectures, such as Transformer-based or structured reasoning models, and on integrating causal reasoning frameworks to enable principled identification of task-critical semantic variables. Extending the proposed framework to more realistic wireless channel models and deploying it on resource-constrained edge platforms will further facilitate the evaluation of robustness, efficiency, and scalability in practical intelligent communication scenarios.

REFERENCES

- [1] T. M. Getu, G. Kaddoum, and M. Bennis, "Semantic communication: A survey on research landscape, challenges, and future directions," *Proceedings of the IEEE*, vol. 112, no. 11, pp. 1649–1685, 2024.
- [2] P. Jonsson, P. Cerwall, A. Lundvall, M. Ekstrand, D. von Koch, F. Fornstad, M. Arvedson, G. Blennerud, V. Chen, L. E. Taylor, P. Lindberg, P. Linder, T. Lodhi, J. Travers, and J. Yazlle, "Ericsson Mobility Report," Ericsson, Technical Report, Jun. 2023.
- [3] Z. Lu, R. Li, K. Lu, X. Chen, E. Hossain, Z. Zhao, and H. Zhang, "Semantics-empowered communications: A tutorial-cum-survey," *IEEE Communications Surveys Tutorials*, vol. 26, no. 1, pp. 41–79, 2024.
- [4] P. A. Stavrou and M. Kountouris, "The role of fidelity in goal-oriented semantic communication: A rate distortion approach," *IEEE Transactions on Communications*, vol. 71, no. 7, pp. 3918–3931, 2023.
- [5] L. Hu, L. Yu, and Z. Qin, "Deep learning-based semantic communication system for wireless image transmission," *IEEE Wireless Communications Letters*, vol. 14, no. 8, pp. 2391–2395, 2025.
- [6] F. Minani, M. Kobayashi, T. Fujihashi, M. A. Alim, S. Saruwatari, M. Nishi, and T. Watanabe, "Resource allocation for semantic communication in ntn downlinks," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 524–544, 2026.
- [7] Z. Sun, S. Ma, and S. Li, "Task-oriented semantic communication with importance-aware rate control," *IEEE Communications Letters*, vol. 29, no. 7, pp. 1520–1524, 2025.
- [8] M. Tao, J. Fan, J. Luo, and H. Xie, "Coarse-to-fine semantic communication systems for text transmission," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 8, pp. 13 267–13 271, 2025.
- [9] T. Han, Q. Yang, Z. Shi, S. He, and Z. Zhang, "Semantic-preserved communication system for highly efficient speech transmission," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 1, pp. 245–259, 2023.
- [10] Y. Wang, C. Dong, W. An, Z. Bao, H. Jiang, Y. Sun, M. Sun, and X. Xu, "Coverage-enhanced semantic communication systems for cellular networks," *IEEE Transactions on Communications*, pp. 1–1, 2026.
- [11] G. Ding, H. Gao, S. Liu, and G. Yu, "Multi-stage semantic communication for low-latency edge inference," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 267–282, 2026.
- [12] Z. Zhang, Q. Yang, S. He, and Z. Shi, "Bi-directional motion enhanced semantic communication for wireless video transmission," *IEEE Internet of Things Journal*, pp. 1–1, 2026.
- [13] H. Gao, M. Sun, X. Xu, X. Cheng, S. Han, and P. Zhang, "Adaptive cross-modal super-resolution semantic communication for mobile ai-

- generated panoramic video,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 427–442, 2026.
- [14] T. M. Getu, G. Kaddoum, and M. Bennis, “Making sense of meaning: A survey on metrics for semantic and goal-oriented communication,” *IEEE Access*, vol. 11, pp. 45 456–45 492, 2023.
 - [15] B. Tang, Q. Li, L. Huang, and Y. Yin, “Text semantic communication systems with sentence-level semantic fidelity,” in *2023 IEEE Wireless Communications and Networking Conference (WCNC)*, 2023, pp. 1–6.
 - [16] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, “Deep learning enabled semantic communication systems,” *IEEE Transactions on Signal Processing*, vol. 69, pp. 2663–2675, 2021.
 - [17] D. Wheeler and B. Natarajan, “Conceptual learning and causal reasoning for semantic communication,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 6, pp. 3669–3683, 2025.
 - [18] S. Ma, W. Qiao, Y. Wu, H. Li, G. Shi, D. Gao, Y. Shi, S. Li, and N. Al-Dhahir, “Task-oriented explainable semantic communications,” *IEEE Transactions on Wireless Communications*, vol. 22, no. 12, pp. 9248–9262, 2023.
 - [19] D. Wheeler and B. Natarajan, “Knowledge-Driven Semantic Communication Enabled by the Geometry of Meaning,” arXiv preprint arXiv:2306.02917, 2023, available online.
 - [20] D. Wheeler, E. E. Tripp, and B. Natarajan, “Semantic communication with conceptual spaces,” *IEEE Communications Letters*, vol. 27, no. 2, pp. 532–535, 2023.
 - [21] D. Wheeler and B. Natarajan, “Autoencoder-based domain learning for semantic communication with conceptual spaces,” in *2024 Wireless Telecommunications Symposium (WTS)*, 2024, pp. 1–6.
 - [22] H. Wan, Q. Yang, J. Tang, and Z. Shi, “Knowledge-aided semantic communication leveraging probabilistic graphical modeling,” in *2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*, 2024, pp. 1–5.
 - [23] F. Zhao, Y. Sun, L. Feng, L. Zhang, and D. Zhao, “Enhancing reasoning ability in semantic communication through generative ai-assisted knowledge construction,” *IEEE Communications Letters*, vol. 28, no. 4, pp. 832–836, 2024.
 - [24] H. Seo, J. Park, M. Bennis, and M. Debbah, “Semantics-native communication via contextual reasoning,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 9, no. 3, pp. 604–617, 2023.
 - [25] Y. Xiao, Y. Liao, Y. Li, G. Shi, H. V. Poor, W. Saad, M. Debbah, and M. Bennis, “Reasoning over the air: A reasoning- based implicit semantic-aware communication framework,” *IEEE Transactions on Wireless Communications*, vol. 23, no. 4, pp. 3839–3855, 2024.
 - [26] C. Xing, J. Lv, T. Luo, and Z. Zhang, “Representation and fusion based on knowledge graph in multi-modal semantic communication,” *IEEE Wireless Communications Letters*, vol. 13, no. 5, pp. 1344–1348, 2024.
 - [27] C. Kurisummoottil Thomas, W. Saad, and Y. Xiao, “Causal semantic communication for digital twins: A generalizable imitation learning approach,” *IEEE Journal on Selected Areas in Information Theory*, vol. 4, pp. 698–717, 2023.
 - [28] C. K. Thomas, C. Chaccour, W. Saad, M. Debbah, and C. S. Hong, “Causal reasoning: Charting a revolutionary course for next-generation ai-native wireless networks,” *IEEE Vehicular Technology Magazine*, vol. 19, no. 1, pp. 16–31, 2024.
 - [29] K.-H. Lee, H.-H. Choi, and J.-R. Lee, “Attention-driven interference alignment for multi-user semantic communications,” *IEEE Transactions on Vehicular Technology*, pp. 1–6, 2025.
 - [30] —, “Attention-driven semantic transmission scheme for ai-native wireless communications,” *IEEE Communications Letters*, vol. 30, pp. 287–291, 2026.